

ETHAN (YUXUAN) XU

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SKILLS

Programming Languages: Python, R, MATLAB, SQL, C, JavaScript, HTML5, CSS3, LaTeX, Java

ML / Data Science: PyTorch, TensorFlow / Keras, Scikit-learn, NumPy, Pandas, SciPy, Matplotlib

Engineering & Tools: Git, GitHub, Jupyter Notebook, Linux, VS Code, Excel, Docker

Research: Overleaf, Zotero, Google Scholar, arXiv, OSF, Obsidian

EDUCATION

Georgia Institute of Technology, College of Computing, Atlanta, GA Sep 2025 – Jun 2027

M.Sc. in Computer Science | Artificial Intelligence Specialization GPA: 4.0

University of Colorado Boulder, College of Engineering & Applied Science, Boulder, CO Jun 2024 – Aug 2024

Graduate-level Coursework (Non-Degree): Algorithms, Advanced Data Structures, Quantum Computing GPA: 4.0

University of Toronto, St. George Campus, New College, Faculty of Arts and Science Sep 2019 – Jun 2023

H.B.A. | Mathematics Major | Philosophy Specialist

PUBLICATIONS

Rapid Sequential Touches Complicate Whisking-Phase Analysis in Mouse Active Touch 2026

Journal of Neurophysiology. Sole-authored neural spike-train, touch-timing, movement, and phase analysis using held-out predictive evaluation; DOI: 10.1152/jn.00211.2026.

Event Files are Common, But Semantic Event Metadata Remain Uneven in OpenNeuro BIDS Datasets 2026

Neuroinformatics. Sole-authored study of BIDS event-metadata completeness, dataset quality, and reproducible neuroinformatics reporting; DOI: 10.1007/s12021-026-09797-y.

SELECTED PROJECTS

George B. Moody PhysioNet Challenge 2026 | Paper Accepted; Poster Forthcoming Feb 2026–Present

Solo student entry under leaderboard team name “GT NeuroSignals Lab”; built Python/Docker PSG classifiers with subject-level cross-validation and leakage-safe evaluation for cognitive-impairment prediction.

BCI ISLA2026 EEG MI Decoder | Displayed Rank 4/114 | Test Accuracy 0.93 Mar 2026 – May 2026

Built reproducible EEG classification package with MNE preprocessing, stratified CV, CSP-SVM/LDA, Riemannian LR, stacking meta-learner, pytest tests, and per-subject CV reports.

PROFESSIONAL EXPERIENCE

Taacam, Mississauga, ON: Junior Data Scientist | AI/VR Analytics & IP Review Sep 2023 – Feb 2024

Supported patent technical review for AI/VR products through invention summaries, prior-art notes, claim-support materials, commercialization analysis, and Python/SQL product analytics.

LinkedKey, Toronto, ON: Data Analyst | Educational Data Mining & Classification May 2024 – Jun 2024

Built Python/SQL classification pipeline for Waterloo math-contest questions with data cleaning, topic/difficulty labels, automated taxonomy, validation checks, SQL dashboards, and education analytics.

Vineridge Acad. / Hanson Acad. / North Bay HS, Ontario: High School Teacher Mar 2024 – Aug 2025

Taught Grade 10–12 math/statistics with data-informed teaching, assessment metrics, collaboration, progress tracking.

RESEARCH EXPERIENCE

GTEDM Lab, Georgia Tech: Graduate Researcher | Learning Analytics May 2026 – Present

Analyzed 456 CS1301 videos across 26 offerings with Python/Pandas learning analytics, feature extraction, cohort dashboards, data-quality validation, statistical modeling, and model-evaluation reports.

CPSec Lab, Georgia Tech: Graduate Researcher | CPS Data & Program Analysis May 2026 – Present

Built CPS firmware workflow linking ArduPilot SITL traces with Ghidra ARM firmware reverse engineering, Python data extraction, benchmark metrics, and model/program reports.

Hohai University, Nanjing: Research Assistant | Time Series Forecasting May 2023 – Aug 2023

Improved streamflow forecasting 12% with Python time-series ML, cross-validation, and error analysis.